

Kernel machines - PART III

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Interpolation splines: minimum norm interpolation

$$\begin{cases} \min_{f \in \mathcal{H}} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 \\ \text{such that} & f(\mathbf{x}_i) = y_i, \quad i = 1, \dots, n \end{cases}$$

The lagrangian (α_i Lagrange multipliers)

$$L(f, \alpha) = \frac{1}{2} \|f\|^2 - \sum_{i=1}^n \alpha_i (f(\mathbf{x}_i) - y_i)$$

optimality for f : $\nabla_f L(f, \alpha) = 0 \Leftrightarrow f(\mathbf{x}) = \sum_{i=1}^n \alpha_i k(\mathbf{x}_i, \mathbf{x})$

dual formulation (remove f from the lagrangian):

$$Q(\alpha) = -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j k(\mathbf{x}_i, \mathbf{x}_j) + \sum_{i=1}^n \alpha_i y_i \quad \text{solution: } \max_{\alpha \in \mathbb{R}^n} Q(\alpha)$$

$$K\alpha = y$$

Interpolation splines: minimum norm interpolation

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Kernel logistic regression

inspiration: the Bayes rule

$$D(\mathbf{x}) = \text{sign}(f(\mathbf{x}) + \alpha_0) \implies \log\left(\frac{\mathbb{P}(Y=1|\mathbf{x})}{\mathbb{P}(Y=-1|\mathbf{x})}\right) = f(\mathbf{x}) + \alpha_0$$

probabilities:

$$\mathbb{P}(Y = 1|\mathbf{x}) = \frac{\exp^{f(\mathbf{x})+\alpha_0}}{1 + \exp^{f(\mathbf{x})+\alpha_0}} \quad \mathbb{P}(Y = -1|\mathbf{x}) = \frac{1}{1 + \exp^{f(\mathbf{x})+\alpha_0}}$$

Rademacher distribution

$$\mathcal{L}(x_i, y_i, f, \alpha_0) = \mathbb{P}(Y = 1|\mathbf{x}_i)^{\frac{y_i+1}{2}} (1 - \mathbb{P}(Y = 1|\mathbf{x}_i))^{\frac{1-y_i}{2}}$$

penalized likelihood

$$\begin{aligned} J(f, \alpha_0) &= -\sum_{i=1}^n \log(\mathcal{L}(x_i, y_i, f, \alpha_0)) + \frac{\lambda}{2} \|f\|_{\mathcal{H}}^2 \\ &= \sum_{i=1}^n \log\left(1 + \exp^{-y_i(f(\mathbf{x}_i) + \alpha_0)}\right) + \frac{\lambda}{2} \|f\|_{\mathcal{H}}^2 \end{aligned}$$

Kernel logistic regression (2)

$$(\mathcal{R}) \quad \begin{cases} \min_{f \in \mathcal{H}} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 + \frac{1}{\lambda} \sum_{i=1}^n \log(1 + \exp^{-\xi_i}) \\ \text{with} & \xi_i = y_i (f(\mathbf{x}_i) + \alpha_0), \quad i = 1, n \end{cases}$$

Representer theorem

$$J(\alpha, \alpha_0) = \mathbb{I}^\top \log \left(\mathbb{I} + \exp^{\text{diag}(\mathbf{y}) K \alpha + \alpha_0 \mathbf{y}} \right) + \frac{\lambda}{2} \alpha^\top K \alpha$$

gradient vector and Hessian matrix:

$$\nabla_{\alpha} J(\alpha, \alpha_0) = K(\mathbf{y} - (2\mathbf{p} - \mathbb{I})) + \lambda K \alpha$$

$$H_{\alpha} J(\alpha, \alpha_0) = K \text{diag}(\mathbf{p}(\mathbb{I} - \mathbf{p})) K + \lambda K$$

solve the problem using Newton iterations

$$\alpha^{\text{new}} = \alpha^{\text{old}} + (K \text{diag}(\mathbf{p}(\mathbb{I} - \mathbf{p})) K + \lambda K)^{-1} K(\mathbf{y} - (2\mathbf{p} - \mathbb{I}) + \lambda \alpha)$$

Let's summarize

- ▶ pros
 - ▶ Universality
 - ▶ from $\mathcal{H}$ to $\mathbb{R}^n$ using the representer theorem
 - ▶ no (explicit) curse of dimensionality
- ▶ splines $\mathcal{O}(n^3)$ (can be reduced to $\mathcal{O}(n^2)$)
- ▶ logistic regression $\mathcal{O}(kn^3)$ (can be reduced to $\mathcal{O}(kn^2)$)
- ▶ no scalability!

sparseness comes to the rescue!

Roadmap

1 Kernels and the learning problem

- Three learning problems
- Learning from data: the problem
- Kernelizing the linear regression
- Kernel machines: a definition

2 Tools: the functional framework

- In the beginning was the kernel
- Kernel and hypothesis set
- Optimization, loss function and the regularization cost

3 Kernel machines

- Non sparse kernel machines
- sparse kernel machines: SVM
- practical SVM

4 Conclusion

SVM: the separable case (no noise)

$$\left\{ \begin{array}{l} \max_{f, \alpha_0} \quad m \\ \text{with} \quad y_i (f(\mathbf{x}_i) + \alpha_0) \geq m \\ \text{and} \quad \frac{1}{2} \|f\|_{\mathcal{H}}^2 = 1 \end{array} \right\} \Leftrightarrow \left\{ \begin{array}{l} \min_{f, \alpha_0} \quad \frac{1}{2} \|f\|_{\mathcal{H}}^2 \\ \text{with} \quad y_i (f(\mathbf{x}_i) + \alpha_0) \geq 1 \end{array} \right.$$

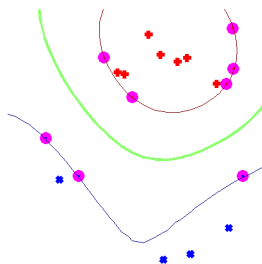
3 ways to represent function f

$$\underbrace{f(\mathbf{x})}_{\text{in the RKHS } \mathcal{H}} = \underbrace{\sum_{j=1}^d w_j \phi_j(\mathbf{x})}_{d \text{ features}} = \underbrace{\sum_{i=1}^n \alpha_i y_i k(\mathbf{x}, \mathbf{x}_i)}_{n \text{ data points}}$$

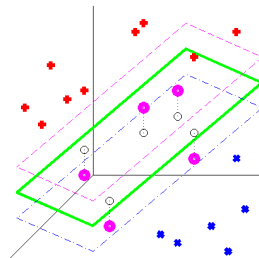
$$\left\{ \begin{array}{l} \min_{\mathbf{w}, \alpha_0} \quad \frac{1}{2} \|\mathbf{w}\|_{\mathbb{R}^d}^2 = \frac{1}{2} \mathbf{w}^\top \mathbf{w} \\ \text{with} \quad y_i (\mathbf{w}^\top \phi(\mathbf{x}_i) + \alpha_0) \geq 1 \end{array} \right\} \Leftrightarrow \left\{ \begin{array}{l} \min_{\boldsymbol{\alpha}, \alpha_0} \quad \frac{1}{2} \boldsymbol{\alpha}^\top K \boldsymbol{\alpha} \\ \text{with} \quad y_i (\boldsymbol{\alpha}^\top K(:, i) + \alpha_0) \geq 1 \end{array} \right.$$

using relevant features...

a data point becomes a function $\mathbf{x} \rightarrow k(\mathbf{x}, \cdot)$



input space representation: $\mathbf{x}$



feature space: $k(\mathbf{x}, \cdot)$

Representer theorem for SVM

$$\begin{cases} \min_{f, \alpha_0} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 \\ \text{with} & y_i (f(\mathbf{x}_i) + \alpha_0) \geq 1 \end{cases}$$

Lagrangian

$$L(f, \alpha_0, \alpha) = \frac{1}{2} \|f\|_{\mathcal{H}}^2 - \sum_{i=1}^n \alpha_i (y_i (f(\mathbf{x}_i) + \alpha_0) - 1) \quad \alpha \geq 0$$

optimality condition: $\nabla_f L(f, \alpha_0, \alpha) = 0 \Leftrightarrow f(\mathbf{x}) = \sum_{i=1}^n \alpha_i y_i k(\mathbf{x}_i, \mathbf{x})$

Eliminate f from L :

$$\begin{cases} \|f\|_{\mathcal{H}}^2 = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j k(\mathbf{x}_i, \mathbf{x}_j) \\ \sum_{i=1}^n \alpha_i y_i f(\mathbf{x}_i) = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j k(\mathbf{x}_i, \mathbf{x}_j) \end{cases}$$

$$Q(\alpha_0, \alpha) = -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j k(\mathbf{x}_i, \mathbf{x}_j) - \sum_{i=1}^n \alpha_i (y_i \alpha_0 - 1)$$

Dual formulation for SVM

the intermediate function

$$Q(\alpha_0, \alpha) = -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j k(\mathbf{x}_i, \mathbf{x}_j) - \alpha_0 \left(\sum_{i=1}^n \alpha_i y_i \right) + \sum_{i=1}^n \alpha_i$$

$$\max_{\alpha} \min_{\alpha_0} Q(\alpha_0, \alpha)$$

α_0 can be seen as the Lagrange multiplier of the following (balanced) constraint $\sum_{i=1}^n \alpha_i y_i = 0$ which is also the optimality KKT condition on α_0

Dual formulation

$$\left\{ \begin{array}{ll} \max_{\alpha \in \mathbf{R}^n} & -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j k(\mathbf{x}_i, \mathbf{x}_j) + \sum_{i=1}^n \alpha_i \\ \text{such that} & \sum_{i=1}^n \alpha_i y_i = 0 \\ \text{and} & 0 \leq \alpha_i, \quad i = 1, n \end{array} \right.$$

SVM dual formulation

Dual formulation

$$\left\{ \begin{array}{l} \max_{\alpha \in \mathbb{R}^n} \quad -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j k(\mathbf{x}_i, \mathbf{x}_j) + \sum_{i=1}^n \alpha_i \\ \text{with} \quad \sum_{i=1}^n \alpha_i y_i = 0 \quad \text{and} \quad 0 \leq \alpha_i, \quad i = 1, n \end{array} \right.$$

The dual formulation gives a quadratic program (QP)

$$\left\{ \begin{array}{l} \min_{\alpha \in \mathbb{R}^n} \quad \frac{1}{2} \alpha^\top G \alpha - \mathbb{I}^\top \alpha \\ \text{with} \quad \alpha^\top \mathbf{y} = 0 \quad \text{and} \quad 0 \leq \alpha \end{array} \right.$$

with $G_{ij} = y_i y_j k(\mathbf{x}_i, \mathbf{x}_j)$

with the linear kernel $f(\mathbf{x}) = \sum_{i=1}^n \alpha_i y_i (\mathbf{x}^\top \mathbf{x}_i) = \sum_{j=1}^d \beta_j x_j$
 when d is small wrt. n primal may be interesting.

Solving the SVM problem

$$\begin{cases} \min_{\alpha \in \mathbb{R}^n} & \frac{1}{2} \alpha^\top G \alpha - \mathbb{I}^\top \alpha \\ \text{with} & \alpha^\top \mathbf{y} = 0 \quad \text{and} \quad 0 \leq \alpha \end{cases}$$

- ▶ assume you know whose $\alpha_i = 0$: $I_0 = \{i \mid \alpha_i = 0\}$
- ▶ $I = \{i \mid \alpha_i \neq 0\}$ (potential support vectors)
- ▶ extract: $\tilde{G} = G(I : I)$, $\tilde{\mathbf{y}} = \mathbf{y}(I)$ and $\tilde{\alpha} = \alpha(I)$

$$\begin{cases} \min_{\alpha \in \mathbb{R}^n} & \frac{1}{2} \tilde{\alpha}^\top \tilde{G} \tilde{\alpha} - \mathbb{I}^\top \tilde{\alpha} \\ \text{with} & \tilde{\alpha}^\top \tilde{\mathbf{y}} = 0 \end{cases} \Rightarrow \begin{aligned} L(\alpha, b) &= \frac{1}{2} \tilde{\alpha}^\top \tilde{G} \tilde{\alpha} - \mathbb{I}^\top \tilde{\alpha} + b \tilde{\mathbf{y}}^\top \tilde{\alpha} \\ \nabla_{\alpha} L(\alpha, b) &= \tilde{G} \tilde{\alpha} - \mathbb{I} + b \tilde{\mathbf{y}} \end{aligned}$$

KKT conditions are:

$$\begin{cases} \nabla_{\alpha} L(\alpha, b) = 0 \\ \tilde{\alpha}^\top \tilde{\mathbf{y}} = 0 \end{cases} \Leftrightarrow \begin{pmatrix} \tilde{G} & \tilde{\mathbf{y}} \\ \tilde{\mathbf{y}}^\top & 0 \end{pmatrix} \begin{pmatrix} \tilde{\alpha} \\ b \end{pmatrix} = \begin{pmatrix} \mathbb{I} \\ 0 \end{pmatrix}$$

Then, if it decreases the cost, move a point: $I \longleftrightarrow I_0$

The importance of being support

$$f(\mathbf{x}) = \sum_{i=1}^n \alpha_i y_i k(\mathbf{x}_i, \mathbf{x})$$

data point	α	constraint value
$\mathbf{x}_i$ useless	$\alpha_i = 0$	$y_i (f(\mathbf{x}_i) + \alpha_0) > 1$
$\mathbf{x}_i$ support	$\alpha_i > 0$	$y_i (f(\mathbf{x}_i) + \alpha_0) = 1$

Table: When a data point is « support » it lies exacty on the margin.

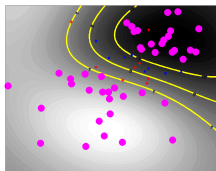
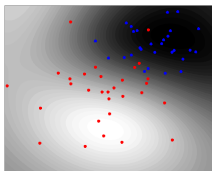
here lies the efficiency of the algorithm (and its complexity)!

sparsness: $\alpha_i = 0$

Data groups: illustration

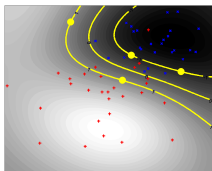
$$f(\mathbf{x}) = \sum_{i=1}^n \alpha_i k(\mathbf{x}, \mathbf{x}_i) + \alpha_0$$

$$D(\mathbf{x}) = \text{sign}(f(\mathbf{x}))$$



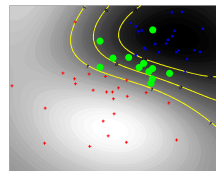
useless data
well classified

$$\alpha = 0$$



important data
support

$$0 < \alpha < C$$



suspicious data

$$\alpha = C$$

the general case: C -SVM

Primal formulation

$$(\mathcal{P}) \left\{ \begin{array}{ll} \min_{f \in \mathcal{H}, \alpha_0, \xi \in \mathbb{R}^n} & \frac{1}{2} \|f\|^2 + \frac{C}{p} \sum_{i=1}^n \xi_i^p \\ \text{such that} & y_i (f(\mathbf{x}_i) + \alpha_0) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, n \end{array} \right.$$

C is the *regularization path* parameter (to be tuned)

$$p = 1, L_1 \text{ SVM} \quad \left\{ \begin{array}{ll} \max_{\alpha \in \mathbb{R}^n} & -\frac{1}{2} \alpha^\top H \alpha + \alpha^\top \mathbb{I} \\ \text{such that} & \alpha^\top \mathbf{y} = 0 \text{ and } 0 \leq \alpha_i \leq C \quad i = 1, n \end{array} \right.$$

$$p = 2, L_2 \text{ SVM} \quad \left\{ \begin{array}{ll} \max_{\alpha \in \mathbb{R}^n} & -\frac{1}{2} \alpha^\top (H + \frac{1}{C} I) \alpha + \alpha^\top \mathbb{I} \\ \text{such that} & \alpha^\top \mathbf{y} = 0 \text{ and } 0 \leq \alpha_i \quad i = 1, n \end{array} \right.$$

the regularization path: is the set of solutions $\alpha(C)$ when C varies

Two more ways to derivate SVM

Using the hinge loss

$$\min_{f \in \mathcal{H}, \alpha_0 \in \mathbb{R}} \frac{1}{p} \sum_{i=1}^n \max(0, 1 - y_i(f(\mathbf{x}_i) + \alpha_0))^p + \frac{1}{2C} \|f\|^2$$

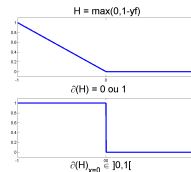
Minimizing the distance between the convex hulls

$$\left\{ \begin{array}{l} \min_{\alpha} \quad \|u - v\|_{\mathcal{H}}^2 \\ \text{with} \quad u(\mathbf{x}) = \sum_{\{i|y_i=1\}} \alpha_i k(\mathbf{x}_i, \mathbf{x}), \quad v(\mathbf{x}) = \sum_{\{i|y_i=-1\}} \alpha_i k(\mathbf{x}_i, \mathbf{x}) \\ \text{and} \quad \sum_{\{i|y_i=1\}} \alpha_i = 1, \quad \sum_{\{i|y_i=-1\}} \alpha_i = 1, \quad 0 \leq \alpha_i \quad i = 1, n \end{array} \right.$$

$$f(\mathbf{x}) = \frac{2}{\|u - v\|_{\mathcal{H}}^2} (u(\mathbf{x}) - v(\mathbf{x})) \quad \text{and} \quad \alpha_0 = \frac{\|u\|_{\mathcal{H}}^2 - \|v\|_{\mathcal{H}}^2}{\|u - v\|_{\mathcal{H}}^2}$$

Regularization path for SVM

$$\min_{f \in \mathcal{H}} \sum_{i=1}^n \max(1 - y_i f(\mathbf{x}_i), 0) + \frac{\lambda}{2} \|f\|_{\mathcal{H}}^2$$



I_α is the set of support vectors s.t. $y_i f(\mathbf{x}_i) = 1$;

$$\nabla_f J(f) = \sum_{i \in I_\alpha} \alpha_i y_i K(\mathbf{x}_i, \cdot) + \sum_{i \in \mathbf{h}} y_i K(\mathbf{x}_i, \cdot) + \lambda f(\cdot) \quad \text{with} \quad \alpha_i = \partial H(\mathbf{x}_i)$$

in particular at point $\mathbf{x}_j \in I_\alpha$ ($f_o(\mathbf{x}_j) = f_n(\mathbf{x}_j) = y_j$)

$$\frac{\sum_{i \in I_\alpha} \alpha_{io} y_i K(\mathbf{x}_i, \mathbf{x}_j) = - \sum_{i \in \mathbf{h}} y_i K(\mathbf{x}_i, \mathbf{x}_j) - \lambda_o y_j}{\sum_{i \in I_\alpha} \alpha_{in} y_i K(\mathbf{x}_i, \mathbf{x}_j) = - \sum_{i \in \mathbf{h}} y_i K(\mathbf{x}_i, \mathbf{x}_j) - \lambda_n y_j}$$

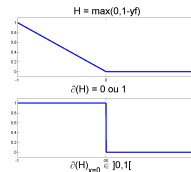
$$G(\alpha_n - \alpha_o) = (\lambda_o - \lambda_n) \mathbf{y} \quad \text{avec} \quad G_{ij} = y_i K(\mathbf{x}_i, \mathbf{x}_j)$$

$$\alpha_n = \alpha_o + (\lambda_o - \lambda_n) \mathbf{w}$$

$$\mathbf{w} = (G)^{-1} \mathbf{y}$$

Regularization path for SVM

$$\min_{f \in \mathcal{H}} \sum_{i=1}^n \max(1 - y_i f(\mathbf{x}_i), 0) + \frac{\lambda}{2} \|f\|_{\mathcal{H}}^2$$



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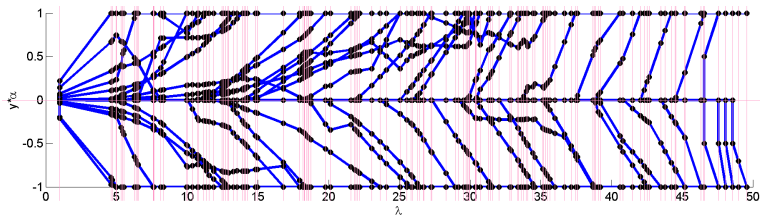
in particular at point $\mathbf{x}_j \in I_\alpha$ ($f_o(\mathbf{x}_j) = f_n(\mathbf{x}_j) = y_j$)

$$\begin{aligned} \sum_{i \in I_\alpha} \alpha_{io} y_i K(\mathbf{x}_i, \mathbf{x}_j) &= - \sum_{i \in \mathbf{h}} y_i K(\mathbf{x}_i, \mathbf{x}_j) - \lambda_o y_j \\ \sum_{i \in I_\alpha} \alpha_{in} y_i K(\mathbf{x}_i, \mathbf{x}_j) &= - \sum_{i \in \mathbf{h}} y_i K(\mathbf{x}_i, \mathbf{x}_j) - \lambda_n y_j \\ \hline G(\alpha_n - \alpha_o) &= (\lambda_o - \lambda_n) \mathbf{y} \quad \text{avec} \quad G_{ij} = y_i K(\mathbf{x}_i, \mathbf{x}_j) \end{aligned}$$

$$\alpha_n = \alpha_o + (\lambda_o - \lambda_n) \mathbf{w}$$

$$\mathbf{w} = (G)^{-1} \mathbf{y}$$

Example of regularization path



α estimation and data selection

How to choose ℓ and P to get linear *regularization path*?

the *path* is piecewise linear $\Leftrightarrow$ one is piecewise quadratic
and the other is piecewise linear

the convex case [Rosset & Zhu, 07]

$$\min_{\beta \in \mathbb{R}^d} \ell(\beta) + \lambda P(\beta)$$

1. piecewise linearity: $\lim_{\varepsilon \rightarrow 0} \frac{\beta(\lambda + \varepsilon) - \beta(\lambda)}{\varepsilon} = \text{constant}$

2. optimality

$$\nabla \ell(\beta(\lambda)) + \lambda \nabla P(\beta(\lambda)) = 0$$

$$\nabla \ell(\beta(\lambda + \varepsilon)) + (\lambda + \varepsilon) \nabla P(\beta(\lambda + \varepsilon)) = 0$$

3. Taylor expansion

$$\lim_{\varepsilon \rightarrow 0} \frac{\beta(\lambda + \varepsilon) - \beta(\lambda)}{\varepsilon} = [\nabla^2 \ell(\beta(\lambda)) + \lambda \nabla^2 P(\beta(\lambda))]^{-1} \nabla P(\beta(\lambda))$$

$$\nabla^2 \ell(\beta(\lambda)) = \text{constant} \quad \text{and} \quad \nabla^2 P(\beta(\lambda)) = 0$$

Problems with Piecewise linear regularization path

L	P	<i>regression</i>	<i>classification</i>	<i>clustering</i>
L_2	L_1	Lasso/LARS	L1 L2 SVM	PCA L1
L_1	L_2	SVR	SVM	OC SVM
L_1	L_1	L1 LAD Danzig Selector	L1 SVM	

Table: example of piecewise linear regularization path algorithms.

$$P : L_p = \sum_{j=1}^d |\beta_j|^p$$

$$L : L_p : |f(\mathbf{x}) - y|^p \quad \text{hinge } (yf(\mathbf{x}) - 1)_+^p$$

$$\varepsilon\text{-insensitive} \quad \begin{cases} 0 & \text{if } |f(\mathbf{x}) - y| < \varepsilon \\ |f(\mathbf{x}) - y| - \varepsilon & \text{else} \end{cases}$$

$$\text{Huber's loss:} \quad \begin{cases} |f(\mathbf{x}) - y|^2 & \text{if } |f(\mathbf{x}) - y| < t \\ 2t|f(\mathbf{x}) - y| - t^2 & \text{else} \end{cases}$$

standart formulation

- ▶ portfolio optimization (Markovitz, 1952)
 - ▶ return vs. risk
$$\begin{cases} \min_{\beta} & \frac{1}{2}\beta^{\top} Q \beta \\ \text{with} & \mathbf{e}^{\top} \beta = C \end{cases}$$



- ▶ *efficiency frontier*: piecewise linear (*Critical path Algo.*)

- ▶ Sensitivity analysis: standart formulation (Heller, 1954)

$$\begin{cases} \min_{\beta} & \frac{1}{2}\beta^{\top} Q \beta + (\mathbf{c} + \lambda \Delta \mathbf{c})^{\top} \beta \\ \text{with} & A\beta = \mathbf{b} + \mu \Delta \mathbf{b} \end{cases}$$

- ▶ Parametric programming (see T. Gal's book 1968)
 - ▶ in the general case of PLP: the reg. path is piecewise linear
 - ▶ ... and PQP is piecewise quadratic
 - ▶ multiparametric programming

ν -SVM and other formulations...

$$\nu \in [0, 1]$$

$$(\nu) \left\{ \begin{array}{ll} \min_{f, \alpha_0, \xi, m} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 + \frac{1}{np} \sum_{i=1}^n \xi_i^p - \nu m \\ \text{with} & y_i (f(\mathbf{x}_i) + \alpha_0) \geq m - \xi_i, \quad i = 1, n, \\ \text{and} & m \geq 0, \quad \xi_i \geq 0, \quad i = 1, n, \end{array} \right.$$

for $p = 1$ the dual formulation is:

$$\left\{ \begin{array}{ll} \max_{\alpha \in \mathbf{R}^n} & -\frac{1}{2} \alpha^\top G \alpha \\ \text{with} & \alpha^\top \mathbf{y} = 0 \text{ et } 0 \leq \alpha_i \leq \frac{1}{n} \quad i = 1, n \\ \text{and} & \nu \leq \alpha^\top \mathbb{I} \end{array} \right.$$

$$C = \frac{1}{m}$$

SVM with non symmetric costs

problem in the primal

$$\left\{ \begin{array}{ll} \min_{f \in \mathcal{H}, \alpha_0, \xi \in \mathbb{R}^n} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 + C^+ \sum_{\{i|y_i=1\}} \xi_i^p + C^- \sum_{\{i|y_i=-1\}} \xi_i^p \\ \text{with} & y_i(f(\mathbf{x}_i) + \alpha_0) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, n \end{array} \right.$$

for $p = 1$ the dual formulation is the following:

$$\begin{cases} \max_{\alpha \in \mathbb{R}^n} & -\frac{1}{2}\alpha^\top G\alpha + \alpha^\top \mathbb{I} \\ \text{with} & \alpha^\top \mathbf{y} = 0 \text{ and } 0 \leq \alpha_i \leq C^+ \text{ or } C^- \quad i = 1, n \end{cases}$$

SVM reduction (reduced set method))

- ▶ objective: compile the model
- ▶ $f(\mathbf{x}) = \sum_{i=1}^{n_s} \alpha_i k(\mathbf{x}_i, \mathbf{x}), n_s \ll n, \quad n_s \text{ too big}$
- ▶ compiled model as the solution of:

$$g(\mathbf{x}) = \sum_{i=1}^{n_c} \beta_i k(\mathbf{z}_i, \mathbf{x}), n_c \ll n_s$$
- ▶ $\beta, \mathbf{z}_i$ and c are tuned by minimizing:

$$\min_{\beta, \mathbf{z}_j} \|g - f\|_H^2$$

where

$$\min_{\beta, \mathbf{z}_i} \|g - f\|_H^2 = \alpha^\top K_x \alpha + \beta^\top K_z \beta - 2\alpha^\top K_{xz} \beta$$

some authors advice $0,03 \leq \frac{n_c}{n_s} \leq 0,1$

- ▶ solve it by using use (stochastic) gradient (its a RBF problem)

SVM and probabilities (1/2)

logistic regression and the import vector machine

- ▶ Logistic regression is NON sparse
- ▶ kernalize it using the *dictionary* strategy
- ▶ Algorithm:
 - ▶ find the solution of the KLR using only a subset $\mathcal{S}$ of the data
 - ▶ build $\mathcal{S}$ iteratively using active constraint approach
- ▶ this trick brings sparsity
- ▶ it estimates probability
- ▶ it can naturally be generalized to the multiclass case
- ▶ efficient when uses:
 - ▶ a few import vectors
 - ▶ component-wise update procedure
- ▶ extention using L_1 KLR

Multiclass SVM

- ▶ one vs all: winner takes all
- ▶ one vs one:
 - ▶ max-wins voting
 - ▶ pairwise coupling: use probability
- ▶ global approach (size $c \times n$),
 - ▶ formal (different variations)

$$\left\{ \begin{array}{ll} \min_{f \in \mathcal{H}, \alpha_0, \xi \in \mathbb{R}^n} & \frac{1}{2} \sum_{\ell=1}^c \|f_\ell\|_{\mathcal{H}}^2 + \frac{C}{p} \sum_{i=1}^n \sum_{\ell=1, \ell \neq y_i}^c \xi_{i\ell}^p \\ \text{with} & y_i(f_{y_i}(\mathbf{x}_i) + b_{y_i} - f_\ell(\mathbf{x}_i) - b_\ell) \geq 1 - \xi_{i\ell} \\ \text{and} & \xi_{i\ell} \geq 0 \text{ for } i = 1, \dots, n; \ell = 1, \dots, c; \ell \neq y_i \end{array} \right.$$

non consistent estimator but practically usefull

- ▶ structured outputs

approach	problem size	number of sub problems
<i>all together</i>	$n \times c$	1
<i>1 vs. all</i>	n	c
<i>1 vs. 1</i>	$\frac{2n}{c}$	$\frac{c(c-1)}{2}$

Roadmap

1 Kernels and the learning problem

- Three learning problems
- Learning from data: the problem
- Kernelizing the linear regression
- Kernel machines: a definition

2 Tools: the functional framework

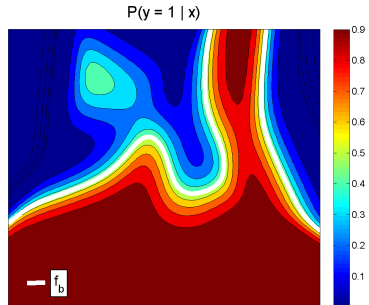
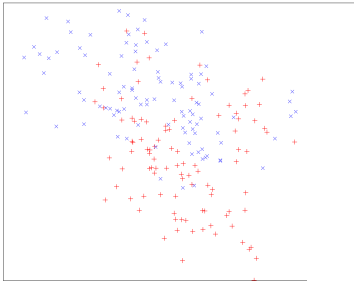
- In the beginning was the kernel
- Kernel and hypothesis set
- Optimization, loss function and the regularization cost

3 Kernel machines

- Non sparse kernel machines
- sparse kernel machines: SVM
- **practical SVM**

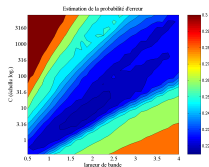
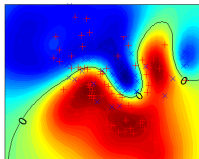
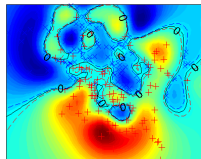
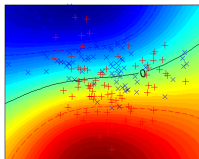
4 Conclusion

Mixture data

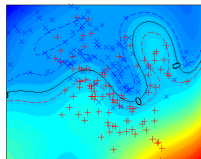


- ▶ $x : 200 \times 2$
- ▶ $y : 100$ for each class
- ▶ mixture model with 10 gaussians
- ▶ the bayes error is known

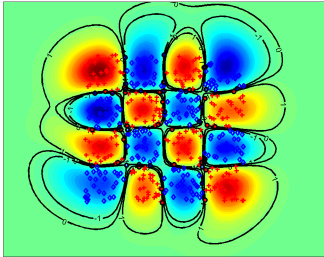
C and σ influence

 $C = 1; \sigma = 1$  $C = 10000; \sigma = 1$  $C = 1; \sigma = 5$ 

$C = 10000; \sigma = 5$

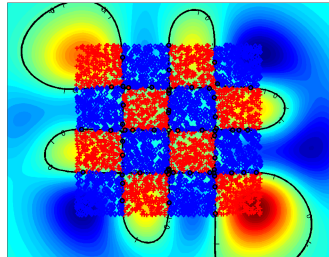


a separable case

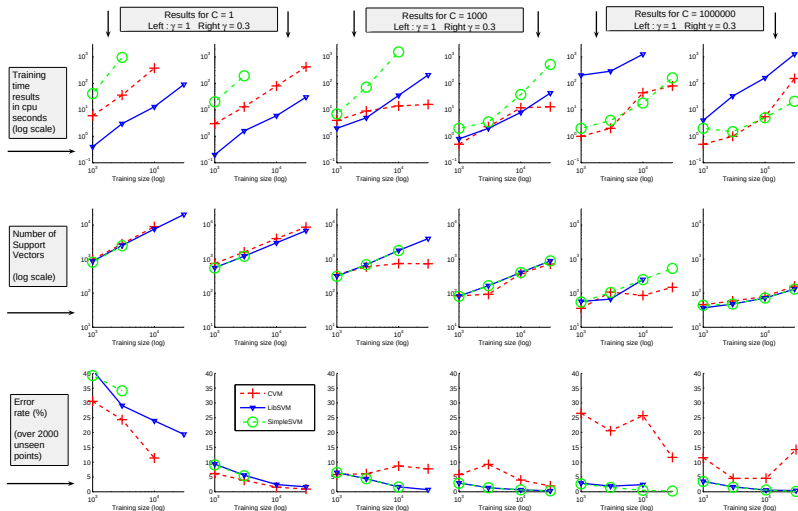


$n = 500$ data points

$n = 5000$ data points



empirical complexity



Conclusion

- ▶ nonlinearity through kernel: using examples influence
- ▶ universality: kernel to functions (R.K.H.S.)
- ▶ representer theorem: from functions to vectors
- ▶ L_1 provides sparsity

a question of vocabulary

- ▶ margin: **regularization**
- ▶ Mercer kernel: **positive kernel**
- ▶ SVM: **a method among others**

- ▶ kernels (RKHS) universality
- ▶ regularization univ. consistency
- ▶ convexity efficiency
- ▶ sparsity efficiency

no (explicit) model

but a kernel, a cost and a regularity

Historical perspective on kernel machines

statistics

1960 Parzen, Nadaraya Watson

1970 Splines

1980 Kernels: Silverman, Hardle...

1990 sparsity: Donoho (pursuit), Tibshirani (Lasso)...

Statistical learning

1985 Neural networks:

- ▶ non linear - universal
- ▶ structural complexity
- ▶ non convex optimization

1992 Vapnik et. al.

- ▶ theory - regularization - consistency
- ▶ convexity - Linearity
- ▶ **Kernel** - universality
- ▶ **sparsity**
- ▶ results: MNIST

